

2520030340

EXPERIMENT – 6

Title:

Process Escape Sequences and Child Process Execution

Aim:

To implement process escape sequences, handle escaped spaces and special symbols, create child processes, execute programs, and test command launching.

Objective:

- To handle escaped spaces.
- To handle escaped special symbols.
- To preserve escaped characters.
- To validate parser output.
- To create child processes.
- To execute programs.
- To pass arguments to programs.
- To handle execution errors.
- To manage the parent process.

Theory:

An escape sequence allows special characters to be treated as normal characters.

For example:

```
echo hello\ world
```

The escaped space is treated as part of the same argument.

A shell can also create a **child process** using `fork()`. The child process can execute a program using `execvp()`, while the parent process can wait for the child using `wait()`.

Algorithm:

1. Start the shell.
2. Display the shell prompt.

3. Read the command from the user.
4. Check for escape characters.
5. Preserve escaped spaces and special symbols.
6. Parse the command into arguments.
7. Create a child process using `fork()`.
8. Execute the command using `execvp()`.
9. Display an error if execution fails.
10. Parent process waits for the child process.
11. Repeat for the next command.
12. Stop when the user enters `exit`.

Functions Used:

Function	Purpose
<code>fgets()</code>	Read command
<code>fork()</code>	Create child process
<code>execvp()</code>	Execute program
<code>wait()</code>	Wait for child process
<code>printf()</code>	Display output
<code>perror()</code>	Display execution errors
<code>exit()</code>	Terminate process

Sample Input:

```
2520030340_kruthi> echo hello\ world
```

Sample Output:

```
hello world
```

Observation:

1. Escaped spaces were handled correctly.
2. Escaped special characters were preserved.
3. Child processes were created successfully.

4. Commands were executed using the child process.
5. Arguments were passed correctly.
6. Execution errors were handled.
7. The parent process waited for the child process.

Result:

The shell successfully handled escape sequences and created child processes to execute commands.

Conclusion:

The experiment successfully demonstrated **escape sequence handling and process creation** using `fork()`, `execvp()`, and `wait()`.

```
kruthi@LAPTOP-LP9QU3RR: ~  
kruthi@LAPTOP-LP9QU3RR:~$ ./shell  
2520030340_kruthi> echo 'hello world'  
hello world  
2520030340_kruthi> echo 'hello    world'  
hello    world  
2520030340_kruthi> echo '$HOME'  
$HOME  
2520030340_kruthi> echo "hello world"  
hello world  
2520030340_kruthi> echo "$HOME"  
/home/kruthi  
2520030340_kruthi> echo ''  
  
2520030340_kruthi> echo ""  
  
2520030340_kruthi> echo 'hello world'  
hello world  
2520030340_kruthi> echo "hello world"  
hello world  
2520030340_kruthi> echo 'hello  
Syntax error: unmatched quote  
2520030340_kruthi> echo "hello  
Syntax error: unmatched quote  
2520030340_kruthi> echo hello  
hello  
2520030340_kruthi> echo hello  
hello  
2520030340_kruthi> echo hello\ \ world  
hello\ \ world  
2520030340_kruthi> echo hello\ \ world  
hello\ \ world  
2520030340_kruthi> echo hello\  
hello\  
2520030340_kruthi> pwd  
/home/kruthi  
2520030340_kruthi> ls  
OSSP                                program.c  
Prime-oss-os                        shell  
directory_name                      shell.c  
os-operating-system-and-system-programming-OSSP  student.txt  
process.c.save                     text.txt  
program  
2520030340_kruthi> echo one two three  
one two three  
2520030340_kruthi> xyzabc  
Command not found: No such file or directory  
2520030340_kruthi> date  
Mon Sep 21 15:44:32 UTC 2026  
2520030340_kruthi> |
```

Output: Escape sequences, escaped spaces, special symbols, command execution, argument passing, and execution error handling.

kruthi@LAPTOP-LP9QU3RR: ~

```
hello
2520030340_kruthi> echo hello
hello
2520030340_kruthi> echo hello\ \ world
hello\ \ world
2520030340_kruthi> echo hello\ \ world
hello\ \ world
2520030340_kruthi> echo hello\!
hello\!
2520030340_kruthi> pwd
/home/kruthi
2520030340_kruthi> ls
OSSP
Prime-oss-p-os
directory_name
os-operating-system-and-system-programming-OSSP
process.c.save
program
program.c
shell
shell.c
student.txt
text.txt
```

```
2520030340_kruthi> echo one two three
one two three
2520030340_kruthi> xyzabc
Command not found: No such file or directory
2520030340_kruthi> date
Mon Sep 21 15:44:32 UTC 2026
2520030340_kruthi> nano shell.c
2520030340_kruthi> nano shell.c
2520030340_kruthi> gcc shell.c -o shell
2520030340_kruthi> ./shell
2520030340_kruthi> ./shell
2520030340_kruthi> echo hello\ world
hello world
2520030340_kruthi> echo hello\!
hello!
2520030340_kruthi> echo hello\!
hello!
2520030340_kruthi> echo one\ two\ three
one two three
2520030340_kruthi> pwd
/home/kruthi
2520030340_kruthi> ls
```

```
OSSP
Prime-oss-p-os
directory_name
os-operating-system-and-system-programming-OSSP
2520030340_kruthi> date
Mon Sep 21 15:50:11 UTC 2026
2520030340_kruthi> xyzabc
Command not found: No such file or directory
2520030340_kruthi> |
process.c.save
program
program.c
shell
shell.c
student.txt
text.txt
```